

MAT 4353 001; MAT 5123 001

Mathematical Foundations of

Cryptography ; Introduction to Cryptography

Course Syllabus - Spring 2025



Course Information

Course Description: Cross-listed courses include MAT 4353.001, MAT 5123.001

Credit Hours: 3; 3

Course Modality: Traditional in-person

Meeting Times

Duration: 01/21/2025-05/16/2025; 01/21/2025-05/16/2025

Campus: Main Campus; Main Campus

Location: [MH 2.01.04](#); [MH 2.01.04](#)

Time(s): MW 4:00 PM - 5:15 PM; MW 4:00 PM - 5:15 PM

Learning Goals

At the end of the course, you will be able to:

- Define the notions of group, cyclic group, subgroup, normal subgroup, homomorphism, isomorphism, providing examples and counterexamples of each. - Define symmetric and permutation groups, express finite permutations as products of cycles and determine their parity, and prove that any group is isomorphic to a permutation group (Cayley's Theorem). - Define the notions of ring, subring, domain, field, ideal, homomorphism, isomorphism, providing examples and counterexamples of each. - Define rings of polynomials, the notion of degree, and irreducible factors. - Use the notions of divisibility, unit and irreducibles to study factorization in rings including $\mathbb{Z}$, $\mathbb{Z}/n$ and $\mathbb{F}[x]$. - Apply concepts and theorems in group, ring and field theory to solve conceptual problems involving specific algebraic structures. - Communicate technical ideas clearly, precisely and concisely in writing.

Communicate with Me

Instructor Name: eduardo.duenez@utsa.edu Eduardo Duenez

Department

Mathematics

Office Location

FLN 4.01.11

Student Hours

TWR 3:00–4:00PM

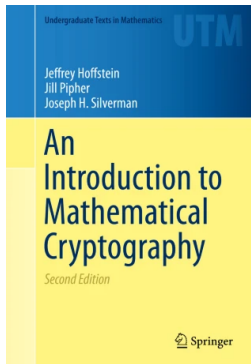
Email Address: eduardo.duenez@utsa.edu**Preferred Method of Communication**

Email

About Me & My Teaching Philosophy

My name is Dr. Eduardo Dueñez. I am an associate professor at the UTSA Mathematics Dept. You can learn more about my academic trajectory and my teaching at [my shortbio](#).

Course Materials

**An Introduction to Mathematical Cryptography****ISBN:** ISBN 978-1-4939-1710-5**Authors:** J. Hoffstein, J. Pipher, J. H. Silverman.**Publisher:** Springer**Publication Date:** 2014**Edition:** 2nd**Required/Recommended/Optional:** Required**Additional Information**

eBook available for download (UTSA Library login required):

<https://link.springer.com/book/10.1007/978-1-4939-1711-2>

Additional Course Information

****Student Ombudsperson:**** Should you have a complaint about this course that cannot be answered by the instructor, or which pertains to unjust or discriminatory treatment, please contact our third-party department representative, Ilse Rosales Cooper ilse.rosalesmartinez@utsa.edu, who can assist with an investigation and resolution of the matter.

Assessments and Assignments

**** Problem sets**

A problem set will be due every week or two.

Submissions should be typeset using $\LaTeX$ or similar specialized software (TeXmacs, LyX).

Work prepared with non-specialized software (e.g., Word) will be accepted, but subject to a 10% penalty.

/Handwritten submissions will not be accepted./

All submissions are to be handed in **hardcopy** by the specified deadline;
no extensions.

** Midterm exams

(There is no final exam for this course.)

- First midterm: Wed 26 February.
- Second midterm: Wed 7 May.

Activities and Grading

** Weight of course activities

- Homework: 30%.
- First midterm exam: 30%.
- Second midterm exam: 40%.

** Final grade ranks

A: [90%, 100%]	B+: [80%, 85%)	C+: [69%, 72%)	D: [50%, 60%)	F: [0%, 50%)
A-: [85%, 90%)	B: [76%, 80%)	C: [65%, 69%)		
	B-: [72%, 76%)	C-: [60%, 65%)		

** Notes

- The lowest (or missing) **single** grade among assignments will be dropped in computing the homework average.
- There is no “make-up” provision to redo any for-credit course activities.
- /No extensions/ will be granted (barring specific circumstances dictated by University policy).

** Time commitment expectations

MAT 4353 is an upper-level /writing-intensive/ course.

The expected /average/ time commitment to this course (which varies week by week) is about **12* hours per week.*

- **Attend** class meetings (**3* hours per week*).
- Class preparation and reading the assigned sections of the textbook (**2–3* hours per week*).
- Solving homework and writing (**3–4* hours per week*).
- Meet and discuss class material with your peer study group (**1–2* hours per week*).
- Attend office hours (**1–2* hours per week*).

Midterm exams are extended take-home tasks. Each may be expected to **take over a dozen hours** over several week- and weekend days.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A	90+
A-	85+
B+	80+
B	76+
B-	72+
C+	69+
C	65+
C-	60+
D	50+
F	<50

Guidelines for the Use of Generative Artificial Intelligence

All uses of Generative Artificial Intelligence are disallowed in this course. Therefore, any use thereof violates the Student Code of Conduct and is subject to applicable University sanctions.

Course Expectations & Policies

** Instructor-Initiated Drops

This course uses instructor-initiated drops for students who exceed the absence and/or missed quiz limit. Up to the last withdrawal day from an individual course (Monday 31 March), students will be dropped for *either* exceeding /three (3) missed graded activities/ *or* /five (5) missed class meetings/. /Attending/ a class meeting is defined as being present for /at least 90% of its duration/. Students will receive at least one courtesy warning when approaching the absence/missed limit. Notification will be sent using ASAP to the student's email address. A subsequent absence or missed assignment will result in being dropped from the course. Notification of being dropped will also be sent through ASAP to the student's email address. This drop does not affect enrollment in other courses. Please consult the Dropping Courses website for further details on the process and appeals.

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable

in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Syllabus Changes

The syllabus is subject to change at the instructor's discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

| Week | Chapters | Topics |

|-----+-----+-----|

| 1/14–1/17 | 1.1 | Substitution cyphers. |

| 1/20–1/24 | 1.2–1.3 | Divisibility and congruences. |

| 1/27–1/31 | 1.4–1.5 | Finite fields $F(p)$. Primitive roots. |

| 2/3–2/7 | 1.7 & | Symmetric vs. Asymmetric Cyphers. |

| | 2.1–2.3 | Discrete Logarithms and the Diffie-Hellman Key Exchange |

| 2/10–2/14 | 2.4–2.6 | Basic group theory. Elgamal PKC. |

| 2/17–2/21 | 2.8 | Basic ring theory. The Chinese Remainder Theorem. |

| 2/24–2/28 | --- | Review/catch-up. First midterm exam (Thu 27 Feb). |

| 3/3–3/7 | 2.7 & 2.9 | Algorithms of Shanks and Pohlig-Hellman. |

| 3/10–3/14 | --- | Spring break. |

| 3/17–3/21 | 2.10 | Polynomial Rings and Finite fields $F(p^k)$. |

| 3/24–3/28 | 3.1–3.3 | Group of units $U(pq)$. The RSA cryptosystem. |

| 3/31–4/4 | 3.4–3.5 | Basic primality testing. Pollard's $p-1$ algorithm. |

| 4/7–4/11 | 4.1–4.3 | Digital signatures (RSA, Elgamal, DSA). |

| 4/14–4/18 | 6.1–6.2 | Introduction to elliptic curves over finite fields. |

| 4/21–4/25 | 6.3–6.5 | Elliptic curves: Discrete logarithms, |

| | Diffie-Hellman, Elgamal, ECDSA signatures. |

| 4/28–5/2 | 26–27 | Lenstra's EC factorization algorithm. Primality certification. |

| 5/5–5/9 | --- | Wrap-up and review. Second midterm exam (Thu 8 May). |

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.

- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UTSA provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit www.utsa.edu/disability or call (210) 458-4157.
- Students at UTSA are responsible for ensuring their work is consistent with UTSA's standards for academic integrity. Students should review [Section 203 of the UTSA Student Code of Conduct](#) for appropriate standards of academic integrity.
- UTSA provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UTSA Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UTSA.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.